Lecture III - Building Reusable Functions

Programming with Python

Dr. Tobias Vlćek



Checkpoint 1

The quarterly board review. The first 40 minutes are the checkpoint. It starts now.

- **Individual work**: no AI, no neighbors, no messaging
- Slides, the cheatsheet and your own notebooks **may be open**
- The **link is on Moodle**. Open it and start
- ~6 short tasks: write code, trace code, fix a bug, answer a multiple choice
- It sweeps **Sessions I-II**: variables, types, `round` and f-strings, `if`/`elif`, `for` loops

When you're done

Cmd/Ctrl+S, then menu → *Download* → *Download Python code* → upload the `.py` to the "Checkpoint 1" assignment on Moodle, **before the 40 minutes are up**. No retakes: one sitting.

> **Note**
> The live checks under the tasks are **provisional**. The final grading runs on my side. Everything in it was rehearsed in the labs.

EP 3: Copy-Paste Soup

After the board review

On to Episode 3.

Tobi has been "reusing" code the only way he knows how: he pasted the same receipt block **14 times**, once per menu item. A ten-cent price change yesterday cost him a whole afternoon of hunting down copies.

Today the code learns to **reuse itself**. We meet the function, and build a first class.

Writing Functions

The copy-paste pain

Tobi totals each receipt line by hand, the same shape, over and over:

```
print(round(2 * 4.50, 2))    # 2 wraps
print(round(1 * 12.00, 2))   # 1 bowl
print(round(5 * 3.20, 2))    # 5 fries
```

```
9.0
12.0
16.0
```

Same calculation, retyped every time. Change the rule once and you must fix it everywhere. There is a better way.

`def`: name the calculation once

Write the calculation **once**, give it a name, and call it whenever you need it:

```
def line_total(qty, price):     # def NAME(inputs):
    return round(qty * price, 2)  # indented body, hands a value back

print(line_total(2, 4.50))
```

```
9.0
```

`def` starts the definition · the name is `line_total` · the block is **indented** · `return` hands the answer back.

Parameters vs arguments

- **Parameters** are the names in the definition, the function's inputs: `qty`, `price`
- **Arguments** are the actual values you pass when you **call** it: `2`, `4.50`

```
def line_total(qty, price):      # qty and price are PARAMETERS
    return round(qty * price, 2)

print(line_total(3, 3.20))       # 3 and 3.20 are ARGUMENTS
```

```
9.6
```

Same function, different arguments, different result. No retyping.

Predict: which number is the bill?

Tobi wants a **40 % tip on a 10 EUR bill**… no wait, a **10 % tip on 40 EUR**. He types it in a hurry. What prints?

```
def with_tip(bill, percent):
    return round(bill * (1 + percent / 100), 2)
```

```
print(with_tip(10, 40))
```

a) `44.0` b) `14.0` c) `Error`

Predict. Pick a letter, then I reveal the answer.

Answer: position decides, not intent

```
def with_tip(bill, percent):
    return round(bill * (1 + percent / 100), 2)

print(with_tip(10, 40))   # bill=10, percent=40
print(with_tip(40, 10))   # bill=40, percent=10
```

```
14.0
44.0
```

b) `14.0`: arguments land on parameters **by position**. `10` becomes `bill`, `40` becomes `percent`: a 40 % tip on 10 EUR. Python cannot read what Tobi meant, only where he put it.

`return`: hand the value back

`return` sends a value **back to whoever called** the function, so you can store it and use it later:

```
def line_total(qty, price):
    return round(qty * price, 2)

subtotal = line_total(3, 3.20)   # catch what came back
print(subtotal)
print(subtotal + 1.50)           # ...and keep using it
```

```
9.6
11.1
```

Predict: print or return?

`label_price` **prints** but has no `return`. What does the **last line** show?

```
def label_price(price):
    print(f"{price:.2f} EUR")

result = label_price(8.50)
print(result)
```

a) `8.50 EUR` b) `Error` c) `None`

Predict. Pick a letter, then I reveal the answer.

Answer: `print` shows, `return` gives

```
def label_price(price):
    print(f"{price:.2f} EUR")

result = label_price(8.50)   # prints while running...
print(result)                # ...but the value handed back is None
```

```
8.50 EUR
None
```

c) **`None`**: `label_price` prints its line while running, but with no `return` it hands back `None`. That `None` is what lands in `result`. Printing is not returning.

Your turn

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_03_a/

Any questions? Let me know!

Default arguments

A parameter can carry a **default**, used when the caller leaves it out:

```
def service_fee(total, rate=0.05):   # rate defaults to 5 %
    return round(total * rate, 2)

print(service_fee(80))         # default rate → 4.0
print(service_fee(80, 0.10))   # override it → 8.0
```

```
4.0
8.0
```

Defaults let one function cover the common case and the special case.

Functions can call functions

A function can use another function you already wrote (reuse builds on reuse):

```
def bill(qty, price):
    items = line_total(qty, price)        # reuse line_total
    return round(items + service_fee(items), 2)  # ...and service_fee

print(bill(2, 4.50))
```

```
9.45
```

`line_total` and `service_fee` do their jobs; `bill` just orchestrates. That's how small pieces become a program.

Your turn

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_03_b/



Any questions? Let me know!

Scope and Classes

Local names stay local

A parameter is the function's **own name** for the value it was handed. Giving that name a new value inside changes nothing outside:

```
def bump(n):
    n = n + 10       # changes the function's OWN copy of n
    return n

print(bump(3))       # 13, the returned value
```

```
13
```

Names created inside a function live in its **scope**. They vanish when the function ends.

Predict: does `stock` move?

`bump` adds ten to its parameter. We call it, then print `stock`. What does the **last line** print?

```
def bump(n):
    n = n + 10
    return n

stock = 3
bump(stock)
print(stock)
```

a) `3` b) `13` c) `Error`

Predict. Pick a letter, then I reveal the answer.

Answer: outside stays untouched

```
def bump(n):
    n = n + 10
    return n

stock = 3
bump(stock)      # the return value is thrown away
print(stock)     # still 3
```

```
3
```

a) `3`: `bump` changes only its own copy `n`. We never stored what it returned, so `stock` out here never moves. The function cannot reach out and rewrite your variables.

Why that's a feature, not a limit

- A function can't **secretly** change your variables, no spooky action at a distance
- You can call it a hundred times and trust your data stays put
- To use a result, you **catch the return** (`stock = bump(stock)`). Deliberate, visible

Isolation is what makes functions **safe to reuse**. That is the whole point of this episode.

A class bundles data with behavior

Sometimes data and the things you do with it belong **together**. A **class** is a blueprint that bundles both:

- **data**: the facts about one thing (a courier's name, their distance)
- **behavior**: what you can compute from it (their fee)

A **method** is just a function that lives inside a class and can read the object's own data.

`class`, `__init__`, and `self`

```
class Delivery:
    def __init__(self, courier, distance_km):  # runs when you build one
        self.courier = courier          # store data ON the object
        self.distance_km = distance_km
        self.rate_per_km = 1.20         # every delivery carries its rate

    def fee(self):                       # a method: note self
        return round(self.distance_km * self.rate_per_km, 2)
```

`__init__` sets up a new object · `self` is *this* object · attributes are stored on `self` and read back through `self`. `fee()` multiplies **two** stored attributes.

Build one, call its method

Instantiate the class to make an object, then use its data and its methods:

```
trip = Delivery("Nadia", 4)   # build one, __init__ runs

print(trip.courier)           # read stored data
print(trip.fee())             # call the method
```

```
Nadia
4.8
```

`trip` is one `Delivery` object. `trip.fee()` computes from *its own* stored distance: data and behavior, traveling together.

Predict: does the second overwrite?

Two deliveries get built from the same class, one after the other. What does the **last line** print?

```
trip = Delivery("Nadia", 4)
second = Delivery("Omar", 10)

print(trip.fee())
```

a) `12.0` b) `Error` c) `4.8`

Predict. Pick a letter, then I reveal the answer.

Answer: each object, its own data

```
trip = Delivery("Nadia", 4)
second = Delivery("Omar", 10)

print(trip.fee())      # self is trip: 4 km
print(second.fee())    # self is second: 10 km
```

```
4.8
12.0
```

c) `4.8`: `__init__` ran twice and stored the data on **two different objects**. `self` inside `trip.fee()` is `trip`, so it reads Nadia's 4 km, not Omar's 10. Two objects of one class do not share their data.

How far we go with classes

- In the lab you add two methods to a class, then write **one small class** yourself: an `__init__` and one method. That's it
- Classes get tricky fast (inheritance, `self` confusion), and that is **not this course**
- We use them only to *bundle* data with behavior, nothing deeper

> **Tip**
> If the `self` keyword feels odd right now, copy the shape from the worked example. It gets clearer after you have written a few.

Your turn

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_03_c/

Any questions? Let me know!

To the Lab

After the break: the lab

- Head to the lab notebook: Episode 3: The Copy-Paste Soup
- You'll write the fee and tip functions, fix a function that forgets to `return`, give a parameter a default, and build the startup's first `Order` class
- It runs entirely in your browser: no setup, just click and code

> ! **Important**
> **Download your `.py` before you leave.** Closing the tab without downloading loses your work, and downloading is exactly how you just handed in the checkpoint.

Wrap-up

Three things to remember

1. A **function** is written once with `def`, takes **parameters**, and hands a value back with `return`: `print` shows, `return` gives back (no `return` → `None`)
2. Names inside a function are **local**: it can't quietly change your variables, which is exactly what makes it safe to reuse
3. A **class** bundles data (on `self`, via `__init__`) with behavior (methods). You'll write just one small one today

> i **Note**
> **Next time, Episode 4:** the menu outgrows Tobi's seventeen loose variables (`price1`, `price2`, `price_final_FINAL2`), and nobody can find anything. We give the data a shape: lists and dictionaries.

Literature

Books to start with

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Link to free online version
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

> i **Note**
> Nothing new here, but these are still great books to start with!

For more, see the literature list of this course.